

Health Screening and Selection: Evidence from Biennial Subsidies in South Korea

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Research questions

Institutional background

Data and Identification

Results

- Effect on take-up

- Selection into screening

- Effect of screening

Conclusion

Health screening, an important form of preventive care

- Important preventive care for leading causes of death
 - Cardiovascular (22%) + Cancer (20%) + Diabetes (3%) + Stroke (5%)

Health screening, an important form of preventive care

- Important preventive care for leading causes of death
 - Cardiovascular (22%) + Cancer (20%) + Diabetes (3%) + Stroke (5%)
- US spending on 5 major cancer screenings: \$43 billion / year ($\approx$ \$130/person)
([Halpern et al., 2024](#))
- Public policy for health screenings
 - ACA preventive care mandate
 - Government-run public screening programs

Do cancer screenings work?

- Many clinical trials and medical organizations guidelines
([Siu and Force, 2016](#); [Nicholson et al., 2024](#); [Davidson et al., 2021](#); [Krist et al., 2021](#); [Grossman et al., 2018](#); [Curry et al., 2018](#))

Do cancer screenings work?

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- One criticism: screening participants are healthier than nonparticipants
 - Higher socioeconomic status
 - Better health behaviors
 - Lower mortality([Pill et al., 1988](#); [Waller et al., 1990](#); [Bender et al., 2015](#); [Carethers and Doubeni, 2020](#))

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- One criticism: screening participants are healthier than nonparticipants
 - Higher socioeconomic status
 - Better health behaviors
 - Lower mortality([Pill et al., 1988](#); [Waller et al., 1990](#); [Bender et al., 2015](#); [Carethers and Doubeni, 2020](#))
- Potential harms are more common among healthy participants
([Welch et al., 2016](#); [Autier et al., 2017](#); [Kowalski, 2023](#))

How can we target health screenings better?

If health screenings are subsidized, who are the marginal participants?

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If health screenings are subsidized, who are the marginal participants?

- Healthier (low-risk) individuals
 - Marginal benefit of a screening
 - Rationale for cost-sharing

([Arrow, 1963](#); [Pauly, 1968](#); [Zeckhauser, 1970](#); [Newhouse, 1993](#); [Manning and Marquis, 1996](#))

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If health screenings are subsidized, who are the marginal participants?

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([Arrow, 1963](#); [Pauly, 1968](#); [Zeckhauser, 1970](#); [Newhouse, 1993](#); [Manning and Marquis, 1996](#))

- Less healthy (high-risk) individuals

- Financial constraints
- Behavioral hazard (present bias, salience)

([Baicker et al., 2015](#); [Choudhry et al., 2011](#); [Bai et al., 2021](#))

Research setting and questions

- National Health Screening Program in South Korea
 - Nationwide program for all the citizens
 - Blanket subsidies at even-numbered ages
 - Repeated throughout lifetime

Research setting and questions

- National Health Screening Program in South Korea
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 - Repeated throughout lifetime
- Research questions
 1. Who are most responsive to the subsidies?
 2. What are the causal effects of screenings on health outcomes and health care utilization?

Preview of results

1. Strong first stage

- Take-up increases by 16-19%P
- Positive spillovers to unsubsidized screenings & spouses

2. High-risk compliers

- Compliers are from lower socioeconomic backgrounds with poorer health conditions

3. Cancer diagnosis & preventive care use ↑

- Cancer diagnoses increase by 17-19%
- Treatment for cancer precursors and cardiovascular risk factors increases

Contributions to literature

1. Selection into screening

- Selection around age 40 or income cutoffs
 - ⇒ Document selection across various SES using broader variation
- Clinical guideline at age 40
 - ⇒ Document selection in response to clinical guideline and subsidies

(Kim and Lee, 2017; Einav et al., 2020; Kowalski, 2023)

Contributions to literature

1. Selection into screening

- Selection around age 40 or income cutoffs
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(Kim and Lee, 2017; Einav et al., 2020; Kowalski, 2023)

2. Causal effects of screening

- Local effects around age or income cutoffs
 - ⇒ Document heterogeneity across age/income
- Single screening type with aggregate cancer diagnoses
 - ⇒ Document impact of 7 cancer screenings with in-situ diagnoses

(Decker, 2005; Kadiyala and Strumpf, 2012, 2016; Bitler and Carpenter, 2016, 2017; Kim and Lee, 2017; Song and Baicker, 2019; Jones et al., 2019; Van Ourti et al., 2020; Myerson et al., 2020; Sun et al., 2022)

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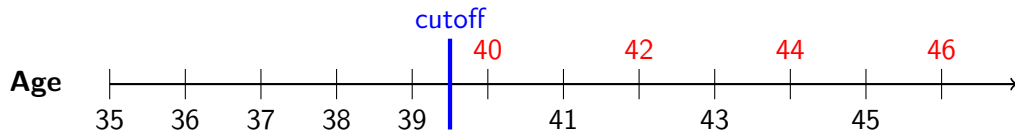
Conclusion

National Health Screening Program

- General health screening
 - Height, weight, waist circumference
 - Sight, hearing
 - Blood pressure, blood sugar and cholesterol level
 - Chest X-ray (tuberculosis), bone density (osteoporosis)
- Cancer screenings

Screening	Tests
Stomach	Upper endoscopy
Breast	Mammography
Cervical	Pap smear
Liver	Alpha-fetoprotein blood test & sonogram
Colorectal	Fecal occult blood test → colonoscopy

Biennial health screening subsidies



- Subsidy amount: 90-100%
 - Total eligible subsidy amount: \$115 (men), \$165 (women)
- ⇒ 5–7% of annual health care expenditure / person

Subsidy schedule

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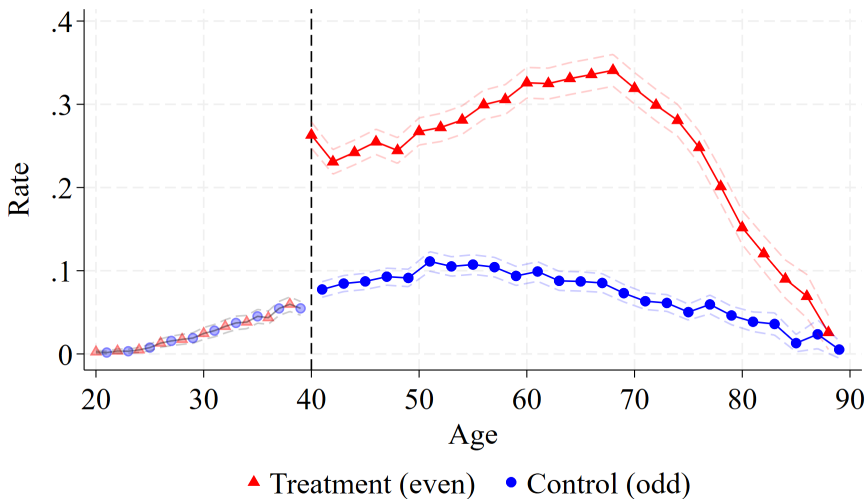
Conclusion

- Korean Health Panel Survey dataset (2008-2018)
 - Analytical sample: 107,200 individual-year
 - Demographics / health care utilization / health behaviors
- Health screening take-up well recorded
 - Date, screening type, tests performed, screening results

Administrative data

- National health insurance claims data (2002-2021)
 - Standard cohort: 8,670,000 individual-year
 - Customized cohort: 7,450,000 individual-year
- Cancer diagnoses available
 - Coinsurance reduction program for rare and severe diseases
 - True positive cancer diagnoses
 - Including cancers diagnosed without a screening
- Incomplete health screening records

Stomach cancer screening take-up by age



Age 40 cutoff variation

- Econometric specification

$$y_{it} = \alpha_0 + \alpha_1 k_{it} + \alpha_2 \mathbb{1}\{k_{it} \geq 0\} + \alpha_3 k_{it} \times \mathbb{1}\{k_{it} \geq 0\} \quad (1)$$

Age 40 cutoff variation

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$$\begin{aligned} y_{it} = & \alpha_0 + \alpha_1 k_{it} + \alpha_2 \mathbb{1}\{k_{it} \geq 0\} + \alpha_3 k_{it} \times \mathbb{1}\{k_{it} \geq 0\} \\ & + treat_{it} \times \{\alpha_4 \mathbb{1}\{k_{it} \geq 0\} + \alpha_5 k_{it} \times \mathbb{1}\{k_{it} \geq 0\}\} \end{aligned} \quad (1)$$

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- Interpretation

$(\hat{\alpha}_2)$	Pre-40	vs	Post-40 control	$\Rightarrow$	clinical guideline
$(\hat{\alpha}_2 + \hat{\alpha}_4)$	Pre-40	vs	Post-40 treatment	$\Rightarrow$	clinical guideline + subsidies
$(\hat{\alpha}_4)$	Post-40 control	vs	Post-40 treatment	$\Rightarrow$	subsidies

Post-40 even ages vs odd ages

- Econometric specification

$$screen_{it} = \beta_0 + \beta_1 \cdot treat_{it} + f(age_{it}) + \epsilon_{it} \quad (2)$$

- Interpretation of even/odd take-up gap $\hat{\beta}_1$

Channel	Interpretation of treatment
Net increase	$\Rightarrow$ Lifetime increase in cancer diagnoses
Intertemporal substitution	$\Rightarrow$ Change in diagnosis timing

Balance tests

- RD analysis at the cutoff
 - Radius: 4 years
 - Balanced across the cutoff Balance
 - Standard error clustered at the individual level
- Even/Odd analysis beyond the cutoff
 - Sample: $\text{Age} \in [40, 89]$
 - Balanced conditional on $f(\text{age})$ Balance
 - Standard error clustered at the individual level

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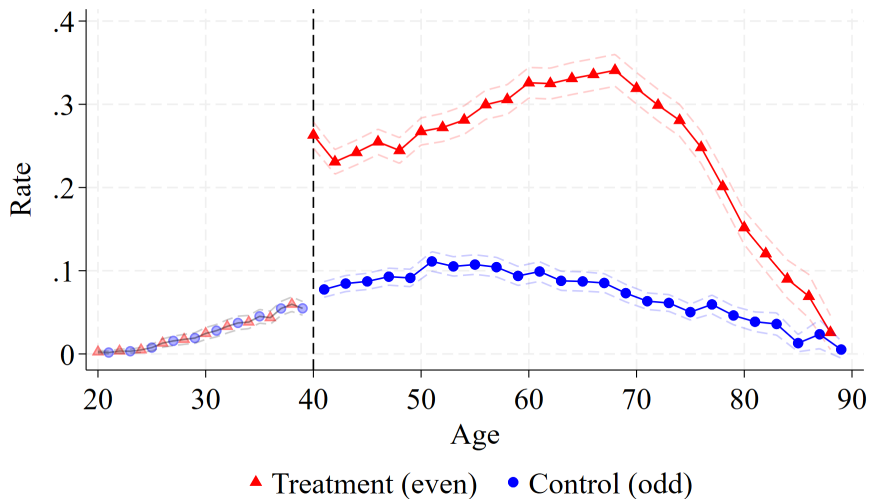
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Stomach screening



RD analyses at the age cutoffs

	(1)	(2)	(3)	(4)
	General	Stomach	Breast	Cervical
Pre-cutoff take-up ($\hat{\alpha}_0$)	0.088*** (0.006)	0.063*** (0.005)	0.056*** (0.007)	0.026*** (0.006)
N	23,623	23,623	12,077	7,797
Sample age range	[36, 43]	[36, 43]	[36, 43]	[26, 33]
Subsidy starting age	40	40	40	30

General

Breast

Cervical

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Change in treatment group ($\hat{\alpha}_2 + \hat{\alpha}_4$)	0.181*** (0.010)	0.201*** (0.009)	0.225*** (0.013)	0.029*** (0.009)
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Post-cutoff treatment – Post-cutoff control ($\hat{\alpha}_4$)	0.173*** (0.011)	0.189*** (0.011)	0.214*** (0.015)	0.024* (0.012)
N	23,623	23,623	12,077	7,797
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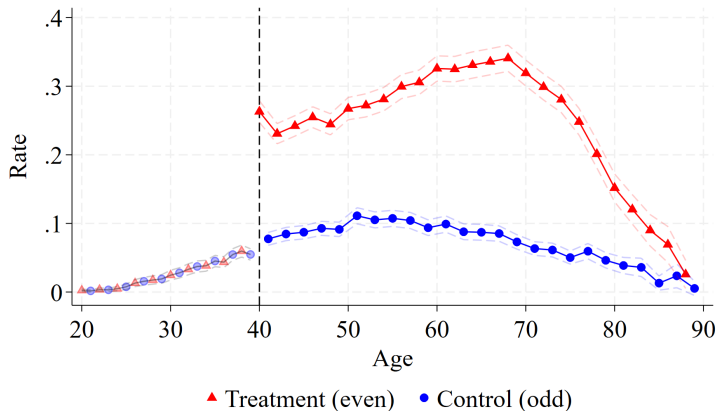
- Effect of clinical guideline + subsidies
- Effect of clinical guideline
- Effect of subsidies

General

Breast

Cervical

Net increase in screening take-up at age 40



- No impact on post-40 control
- Pre-40 and post-40 control are similar
- Minimal substitution from age 39 to 40

AT comparison

Substitution (stomach)

Substitution (breast)

Treatment (even) vs Control (odd) take-up after age 40

	(1)	(2)	(3)	(4)
	General	Stomach	Breast	Cervical
treat	0.187*** (0.003)	0.190*** (0.003)	0.191*** (0.004)	0.164*** (0.003)
N	107,183	107,183	56,923	56,923
Control group mean	0.102	0.083	0.067	0.056
Percentage increase	183	229	283	295

Treatment (even) vs Control (odd) take-up after age 40

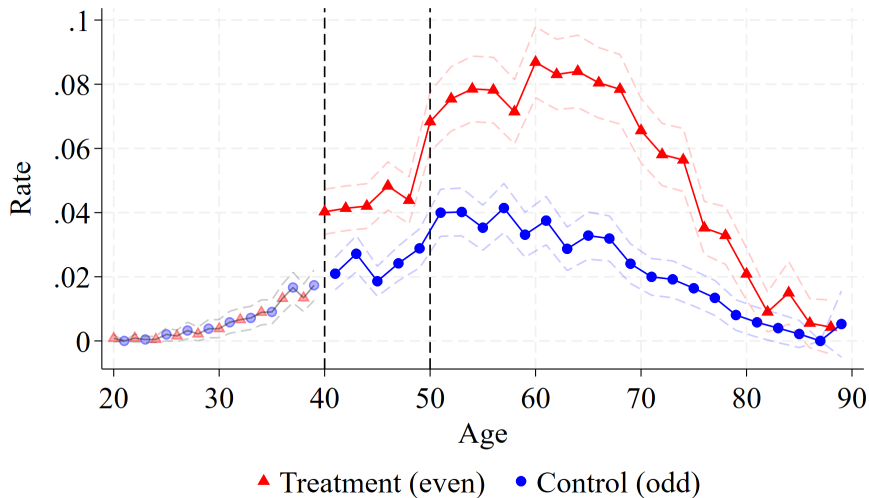
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- Net increase
- Shift in screening timing

Elasticity

Substitution

Colorectal screening - subsidized every year



Cross-screening spillover

	(1)	(2)	(3)	(4)
	Annually subsidized		Unsubsidized	
	Liver	Colorectal	Prostate	Lung
treat	0.027*** (0.001)	0.033*** (0.001)	0.007*** (0.001)	0.0062*** (0.0007)
N	107,183	107,183	50,260	107,183
Control group mean	0.028	0.027	0.009	0.009
Percentage increase	94	124	81	67

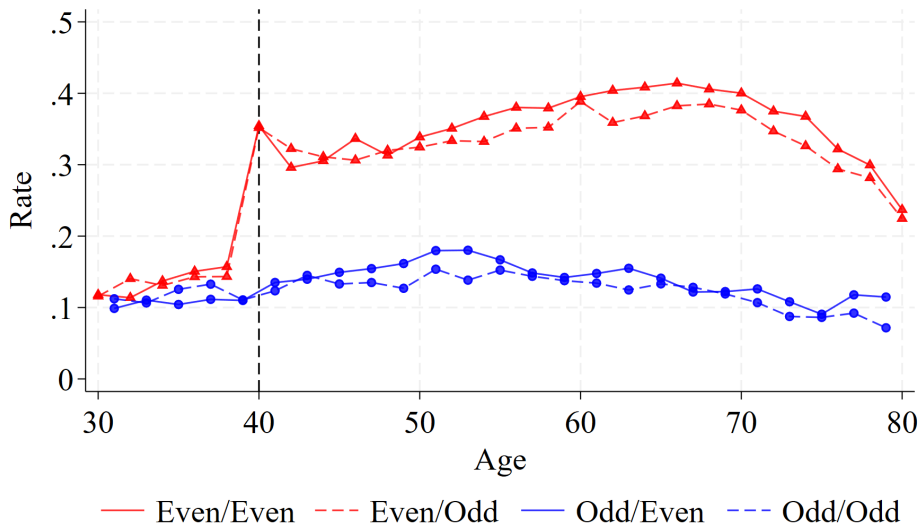
Cross-screening spillover

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N	107,183	107,183	50,260	107,183
Control group mean	0.028	0.027	0.009	0.009
Percentage increase	94	124	81	67

- Receiving multiple screenings during a single hospital visit

Same day share

Spousal spillover in take-up (separate paper)



Spousal spillover in take-up

	(1)	(2)
	Outcome var: Own screening take-up of any kind	
treat	0.217*** (0.006)	0.216*** (0.004)
spouse treat	0.017*** (0.005)	
treat × spouse treat	0.0002 (0.0096)	
spouse screening		0.079*** (0.017)
N	76032	76032
Odd/Odd group mean	0.131	0.131
Estimator	OLS	2SLS

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Estimator	OLS	2SLS

- Peer effect is 37% of the direct subsidy effect
- Husband's take-up is sensitive to wife's take-up, but not vice versa
- Couples receive screenings on the same day

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Characterizing marginal participants (compliers)

- Who are marginal participants?

	Treatment (even)	Control (odd)
Always-takers	Yes	Yes
Compliers	Yes	No
Never-takers	No	No

Characterizing marginal participants (compliers)

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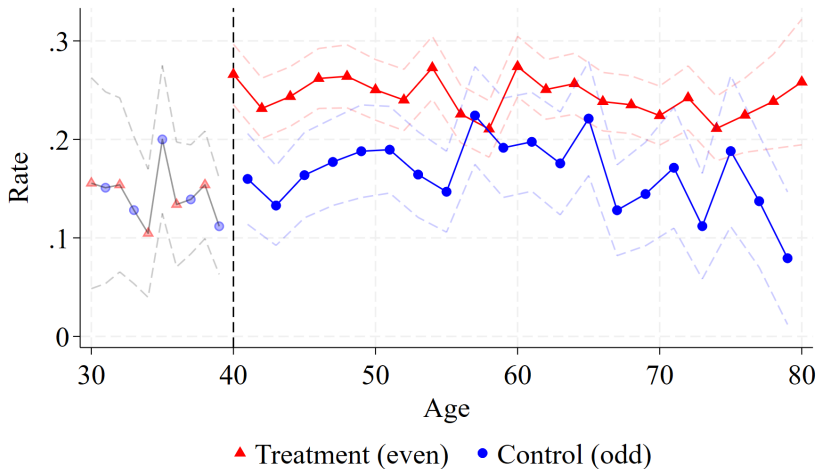
- Cross-sectional approach (within a year)

Participants in control $\Rightarrow$ Always-takers

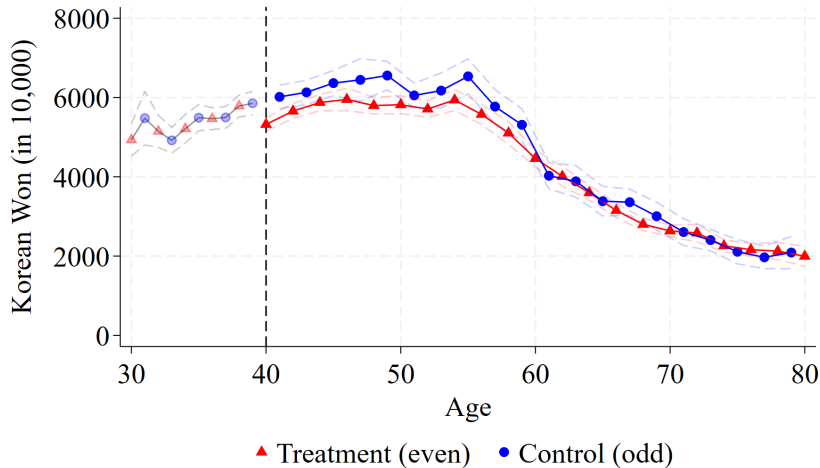
Nonparticipants in treatment $\Rightarrow$ Never-takers

- Panel approach (across 10 years) Panel

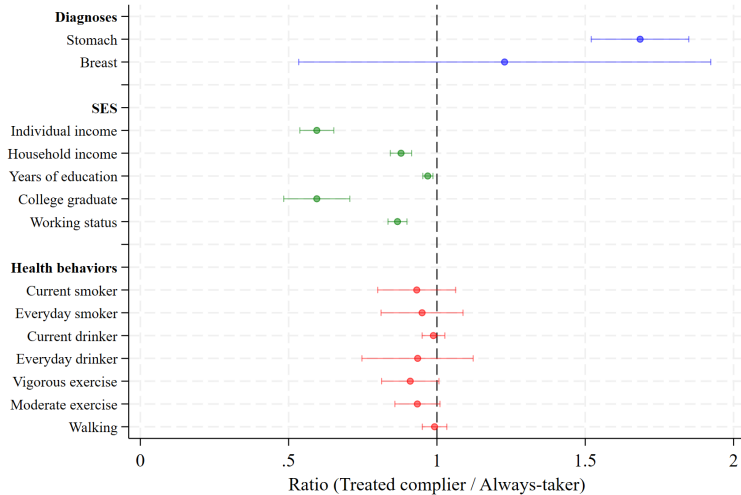
Compliers are more likely to be diagnosed with a stomach disease



Compliers have lower household income



Compliers vs Always-takers at ages [40,89] using even/odd variation



Calculation

Screening results

Cutoff

Stomach diseases

Health behaviors

Non-working

Medical tests

Never

Targeting health screenings toward high-risk group

- Compared to always-takers, compliers have
 - Lower SES
 - Poorer health ([Kim and Lee, 2017](#))

Targeting health screenings toward high-risk group

- Compared to always-takers, compliers have
 - Lower SES
 - Poorer health ([Kim and Lee, 2017](#))
- Subsidies help target high-risk groups
 - Guidelines alone cannot target high-risk (screening, vaccine)
([Einav et al., 2020](#); [Lawler, 2020](#))
 - Lower prices help target high-risk (screening, vaccine, drug adherence)
([Choudhry et al., 2011](#); [Baicker et al., 2015](#); [Kim and Lee, 2017](#); [Chandra et al., 2024](#); [Humlum et al., 2024](#))
 - Administrative burden often excludes low-SES, sicker people (Medicaid, SNAP, SSDI)
([Giannella et al., 2024](#); [Deshpande and Li, 2019](#); [Bhargava and Manoli, 2015](#); [Bettinger et al., 2012](#))

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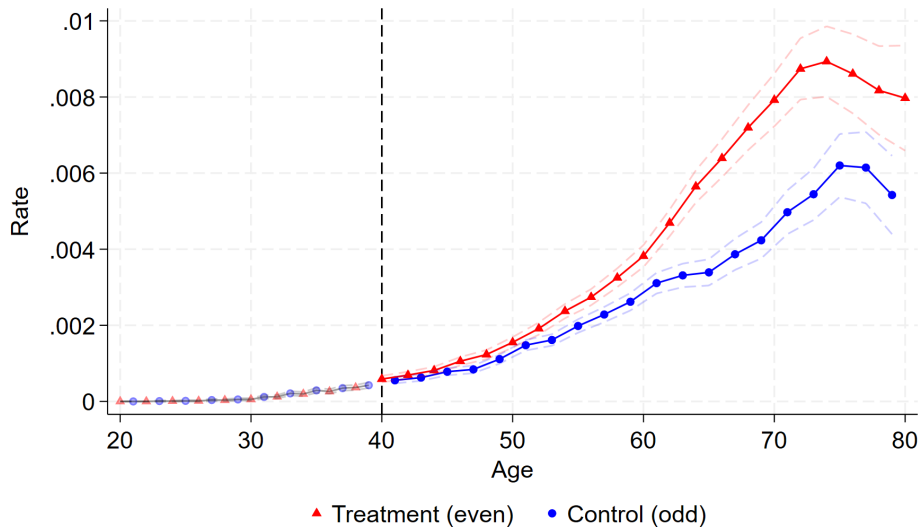
Conclusion

- National Health Insurance claims data
 - All cancer diagnosis (including diagnoses through symptoms)
 - Preventive care use
 - Mortality
- Intent-to-treat (ITT) estimates

$$y_{it} = \eta_0 + \eta_1 \cdot treat_{it} + f(age_{it}) + \varepsilon_{it} \quad (3)$$

- Individual i in year t with age $\in [40, 89]$
- Only capture short-run effect within the same year

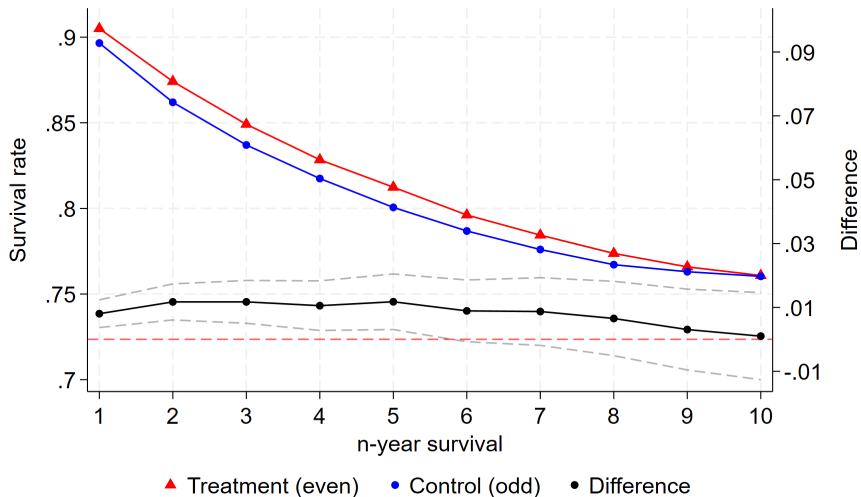
More stomach cancer diagnoses in the treatment group



Screening increases both **in-situ** and **invasive** cancer diagnoses

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
Any cancer	910	160*** (10)	18	7,449,256
invasive	840	140*** (10)	17	7,449,256
in-situ	100	20*** (3)	19	7,449,256
Stomach cancer	190	77*** (6)	41	7,449,256
invasive	180	75*** (5)	41	7,449,256
in-situ	5	2.7*** (0.7)	55	7,449,256
Breast cancer	470	50*** (10)	11	3,503,656
invasive	430	30*** (10)	8	3,503,656
in-situ	70	16*** (4)	24	3,503,656
Cervical cancer	150	20*** (7)	13	3,503,656
invasive	70	8 (5)	11	3,503,656
in-situ	82	14*** (4)	17	3,503,656

Cancer patients diagnosed in the treatment group live longer



Interpreting the causal effects

1. Effect of National Health Screening Program

- Upper bound due to intertemporal substitution,

2. Effect of health screenings

- **Early** diagnosis of cancer

3. Complier cancer risks

- Cancer rate among compliers = LATE

LATE

Screening increases medication use for cardiovascular risk factors

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
High blood pressure	23,884	46*** (13)	0.19	8,673,954
Diabetes	8,918	15** (6)	0.17	8,673,954
High cholesterol	13,901	419*** (11)	3	8,673,954

Precursors

High blood pressure

Diabetes

High cholesterol

Other

Impact on 1-year mortality

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
Total death	908.4	-4.8 (7.2)	-1	6,740,415
Cancer death	280.5	-1.5 (4.1)	-1	6,740,415
Stomach cancer death	33.4	-0.2 (1.4)	-1	6,740,415
Breast cancer death	15.3	-1.1 (1.3)	-7	3,480,294
Cervical cancer death	5.9	0.8 (0.8)	13	3,480,294
Liver cancer death	43.4	-1.5 (1.6)	-4	6,740,415
Colorectal cancer death	28.7	1.6 (1.3)	6	6,740,415
Lung cancer death	64.6	-0.3 (2.0)	-1	6,740,415
Prostate cancer death	11.6	0.3 (1.2)	2	3,260,121

Robustness checks

- Different age controls Balance First Cross Selection
- Bounding estimates with sample adjustment Balance First Cross
- Age 40 cutoff RD analyses First Cross Selection
- Intertemporal substitution in take-up net increase substitution
- Heterogeneity in cancer diagnoses across age, income, disability Heterogeneity

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Takeaways

1. Universal screening program can effectively target high-risk participants
2. Health screenings lead to early detection of diseases and increase preventive care use

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References

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Research question

Institutional background

Identification and data

Effect on take-up

Cross spillover

Selection

Effect

Subsidy schedule across screenings

	Biennial subsidy				Annual subsidy		No subsidy	
	General	Stomach	Breast	Cervical	Liver	Colorectal	Lung	Prostate
Age cutoff	40	40	40	30	40	50		
Subsidy amount	100%	90%	90%	100%	90%	90%	0%	0%
Full cost (\$)	50	65	40	15	90	10	100	20
Target			Female	Female	High risk group			Male

⇒ Total eligible subsidy amount: \$115 (men), \$165 (women)

⇒ 5-7% of annual health care expenditure / person

Liver high risk group

Implementation of the health screening program

- Nationwide program
 - Target: all the citizens covered by National Health Insurance Service (NHIS)
 - History: (1980) Beginning of the program → (2004) Current system
- How can I receive subsidized screenings
 - Providers: public health clinics / private clinics and hospitals designated by the NHIS
⇒ (Dec 2023) 6,600 screening centers for general screening → 4600 people per center
 - Appointment: normally required but varies by hospitals and type of screenings
- Do people know about the screening program and the subsidies?
 - Even-odd subsidy rule has been used throughout the study period
 - Reminder mails (and mobile notifications)
 - Sent to those eligible for subsidies
 - Mail contains the type of screenings to receive and screening providers in the neighborhood

High risk group for liver screening

1. Individuals with the following diseases
 - Cirrhosis
 - Chronic liver disease
2. Individuals who were diagnosed with positive results in the previous year general health screening
 - Hepatitis B surface antigen test
 - Hepatitis C virus HCV antibody test

⇒ can be found through blood test
3. Individuals who used medical services for the following diseases in the past two years are excluded
 - Liver cell carcinoma, hepatocellular carcinoma and liver cancer (C22.0)
 - Intrahepatic bile duct carcinoma and Cholangiocarcinoma (C22.1)

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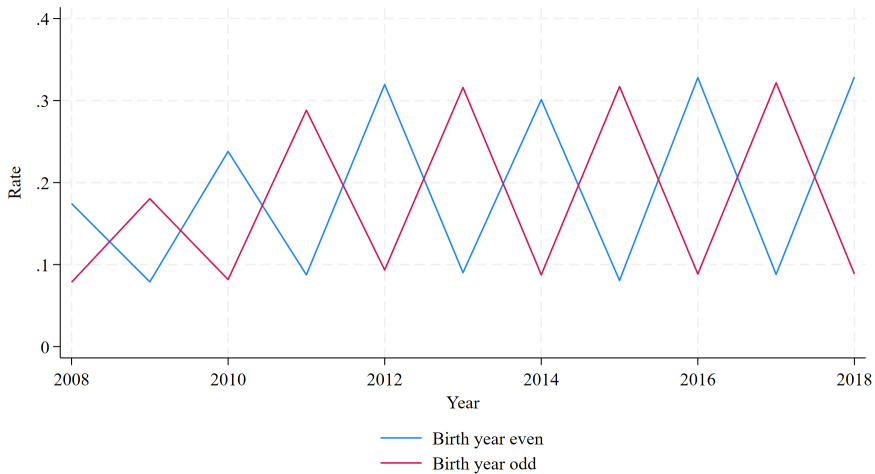
- Effect on take-up

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- Effect

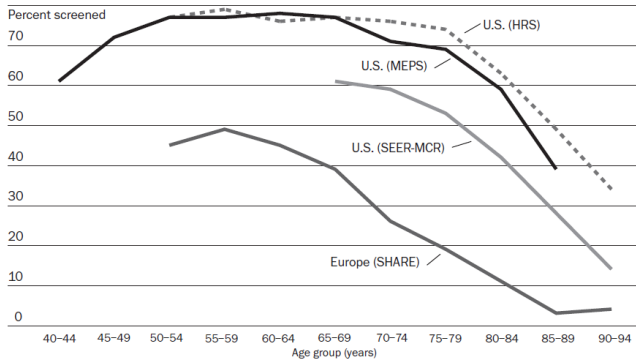
Stomach screening by birth year



Breast screening in the US and Europe (Howard et al., 2009)

EXHIBIT 3

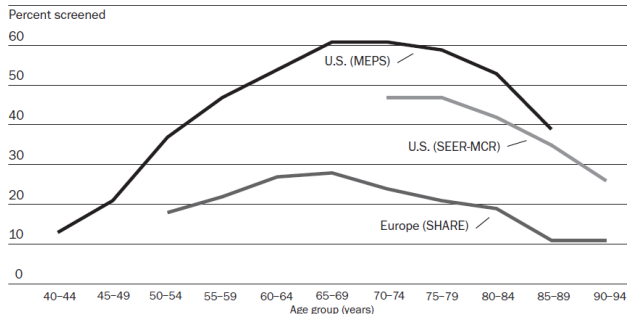
Receipt Of Mammography In The Past Two Years Among Women Ages 44–94 In Europe And The United States, By Age Group, 2004



SOURCES: U.S. screening rates are from the Medical Expenditure Panel Survey (MEPS), the Health and Retirement Study (HRS), and Surveillance, Epidemiology, and End Results (SEER)–Medicare data (SEER-MCR). European rates are from the Survey of Health, Ageing, and Retirement in Europe (SHARE).

Colorectal screening in the US and Europe (Howard et al., 2009)

EXHIBIT 4
Receipt Of Colonoscopy, Sigmoidoscopy, And Fecal Occult Blood Tests Among Women And Men Ages 44–94 In The Past Ten Years In Europe And In The Past Five Years In The United States, By Age Group, 2004



SOURCES: U.S. screening rates are from the Medical Expenditure Panel Survey (MEPS); and Surveillance, Epidemiology, and End Results (SEER)–Medicare data (SEER-MCR). European rates are from the Survey of Health, Ageing, and Retirement in Europe (SHARE).

Balance across the age 40 cutoff

	(1)	(2)	(3)
	Right before 40	Right after 40	Differences
Female	0.503*** (0.011)	0.510*** (0.009)	0.007 (0.007)
Currently married	0.851*** (0.008)	0.845*** (0.007)	-0.006 (0.005)
Years of education	13.707*** (0.052)	13.703*** (0.043)	-0.004 (0.032)
Working status	0.746*** (0.010)	0.752*** (0.008)	0.006 (0.009)
Individual income	2,193*** (47)	2,212*** (40)	19 (35)
Household income	4,852*** (67)	4,922*** (49)	70 (59)
Own a house	0.657*** (0.011)	0.652*** (0.009)	-0.005 (0.009)
Number of household members	3.940*** (0.024)	3.923*** (0.019)	-0.017 (0.016)

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Treatment and control groups are balanced conditional on $f(\text{age})$

	(1)	(2)	(3)
	Treatment (Even)	Control (Odd)	Conditional difference
Age	58.697 (12.532)	59.240 (12.353)	- -
Female	0.530 (0.499)	0.532 (0.499)	-0.002* (0.001)
Currently married	0.799 (0.401)	0.798 (0.402)	-0.0011 (0.0008)
Years of education	10.320 (4.510)	10.227 (4.538)	-0.003 (0.008)
Working status	0.610 (0.488)	0.608 (0.488)	-0.003* (0.001)
Individual income	1446.3 (2081.6)	1425.7 (2068.1)	2.762 (5.185)
Household income	4104.4 (3708.6)	4086.7 (3737.9)	3.221 (14.267)
Own a house	0.734 (0.442)	0.737 (0.441)	-0.0002 (0.0011)
Number of household members	3.067 (1.317)	3.051 (1.317)	-0.004 (0.003)
N	54,274	52,909	
Share	(0.51)	(0.49)	
F(8, 15939)			1.65
Prob > F			(0.10)

Balance table with unconditional differences

	(1)	(2)	(3)	(4)
	Even age group	Odd age group	Unconditional difference	Conditional difference
Age	58.697 (12.532)	59.240 (12.353)	-0.543*** (0.026)	- -
Female	0.530 (0.499)	0.532 (0.499)	-0.002** (0.001)	-0.002* (0.001)
Currently married	0.799 (0.401)	0.798 (0.402)	0.0009 (0.0009)	-0.0011 (0.0008)
Years of education	10.320 (4.510)	10.227 (4.538)	0.093*** (0.009)	-0.003 (0.008)
Working status	0.610 (0.488)	0.608 (0.488)	0.001 (0.002)	-0.003* (0.001)
Individual income	1446.3 (2081.6)	1425.7 (2068.1)	20.607*** (5.508)	2.762 (5.185)
Household income	4104.4 (3708.6)	4086.7 (3737.9)	17.735 (14.555)	3.221 (14.267)
Own a house	0.734 (0.442)	0.737 (0.441)	-0.002* (0.001)	-0.0002 (0.0011)
Number of household members	3.067 (1.317)	3.051 (1.317)	0.016*** (0.003)	-0.004 (0.003)
N	54274	52909		
Share	(0.51)	(0.49)		
F(8, 15939)				1.65 (0.10)

Robustness: balance table with different age controls

	(1)	(2)	(3)
	3 years	5 years	7 years
Female	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)
Currently married	-0.0014* (0.0009)	-0.0011 (0.0008)	-0.0012 (0.0008)
Years of education	-0.003 (0.008)	-0.003 (0.008)	-0.003 (0.008)
Working status	-0.003** (0.002)	-0.003* (0.001)	-0.003* (0.001)
Individual income	1.1 (5.3)	2.8 (5.2)	1.2 (5.2)
Household income	0.6 (15.4)	3.2 (14.3)	-4.5 (14.1)
Own a house	-0.0002 (0.0011)	-0.0002 (0.0011)	0.00002 (0.00109)
Number of household members	-0.004 (0.003)	-0.004 (0.003)	-0.004* (0.003)

Balance

Robustness

Bounding estimates: balance table

	(1)	(2)	(3)
	Age $\in [39, 89]$	Age $\in [40, 89]$	Age $\in [41, 89]$
Age	0.521*** (0.026)	-0.543*** (0.026)	0.562*** (0.025)
Female	-0.001 (0.001)	-0.002** (0.001)	-0.001 (0.001)
Currently married	-0.0009 (0.0009)	0.0009 (0.0009)	-0.0018** (0.0009)
Years of education	-0.094*** (0.009)	0.093*** (0.009)	-0.107*** (0.010)
Working status	-0.006*** (0.001)	0.001 (0.002)	-0.007*** (0.002)
Individual income	-16.235*** (5.470)	20.607*** (5.508)	-24.789*** (5.618)
Household income	-23.153 (14.766)	17.735 (14.555)	-31.182** (14.995)
Own a house	0.003*** (0.001)	-0.002* (0.001)	0.003** (0.001)
Number of household members	-0.027*** (0.003)	0.016*** (0.003)	-0.034*** (0.003)
N	110121	107183	104153

Balance

Robustness

Health care usage data collection

- Recording health care usage
 - Survey participants are asked to keep health diary and store receipts from every visit to hospitals and pharmacies
- No gap
 - During annual interviews, enumerator goes through health diary from the last time of interview

Health diary

● 건강가계부 작성방법 ●

● 병의원에 다녀왔을 때

- ▶ 우리 가족 누구든지 병의원에 다녀오면 가계부를 작성해주세요.
- ▶ 병의원 영수증과 처방전 및 약국 영수증은 영수증 보관함에 함께 모아주세요.

<작성 예시> 아들 홍길동이 이비인후과에 비염 때문에 다녀온 후

의료 이용 형태	☒ 외래 ☐ 입원 ☐ 응급 ☐ 건강검진
진료 일자	2019년 4월 10일 (목) 월 일
가 구 원 이 름	홍길동
병 의 원 이 름	문현한 이비인후과
방 문 이 유	알레르기 비염
병 원 수납 금액	4,000 원
교 통 수 단	내선 ☐ 걸어서 ☐ 주차 ☐ 걸어서
배 스, 택 시, 도 보 등	
보 관 여 부	☒ 진료비 납입 영수증 ☐ 처방전 ☒ 약국영수증

● 의약품 및 보건의료용품을 샀을 때

- ▶ 우리 가족 누구든지 처방전 없이 의약품 또는 의료기기, 건강기능식품 등을 구매하면 가계부에 기입해주세요.
- ▶ 다음과 같은 항목을 구매한 경우 특별로 합산하여 기입해주세요.
※ 구입영수증은 영수증 보관함에 따로 모아주세요.

<예시> 알레르기 항에 먹으려고 알레르기인 구입, 감기 기운이 있어 알레르기 종합감기약을 약국에서 구매

2019년 1월			
구입품목	구입 장소	비율	
1. 일반의약품/의약품	☐ 병원 ☒ 약국 ☐ 인터넷/편의점	(6,000) 원	
2. 한약 및 한약재 (처방 한약 제외)	☐ 약국 ☐ 한약방	() 원	
3. 건강보조식품 (홍삼, 비타민 등)	☐ 병원 및 약국 ☒ 인터넷 및 홈쇼핑 ☐ 백화점, 마트, 시장 등	(47,500) 원	
4. 의료기기 및 의료용품 ※ 예시 - 보건의료수용물 (밴드, 마스크, 시안수, 안대, 모기장(집안 등)) - 안경 및 콘택트렌즈 구입 및 수리 - 보청기 구입 및 수리 - 신장보조용 의료기기 등 기타 의료용품을 구매, 대여 및 수리 (혈액기, 인공 혈관기, 복발, 척추 교정기, 혈당측정기 등)		() 원	

<How to write health diary>

- Visit to hospital
 - Record it for all the household members
 - Store hospital receipts, prescriptions and pharmacy receipts in a box

<Example> After a visit to ENT for allergy

Type	☐ Outpatient ☐ Inpatient ☐ Emergency ☐ Screening			
Date	From: April 10, 2019 To:			
Name	John Doe			
Name of the hospital	Dr. Jane M. Doe, MD			
Purpose	Allergy			
Hospital bills	\$40			
Transportation	To	Walking	From	Walking
Receipts	☐ Hospital ☐ Prescription ☐ Pharmacy			

- Purchase of OTC drugs, oriental medicine, dietary supplements
 - Record it for all the household members
 - Store hospital receipts, prescriptions and pharmacy receipts in a box

<Example> Purchase of multivitamin and Tylenol

January 2019		
Item	Place	Cost
OTC drugs	☐ Hospital ☐ Pharmacy ☐ CVS	{ } KRW { } KRW { } KRW
Oriental medicine	☐ Pharmacy ☐ Acupuncture clinic	{ } KRW { } KRW
Dietary supplement (ginseng, vitamin, etc)	☐ Hospital or pharmacy ☐ Internet shopping ☐ Department store	{ } KRW { } KRW { } KRW
Any other medical products (e.g.) - Bandage, mask, insect repellent - Glasses, contact lenses - Hearing aid		{ } KRW

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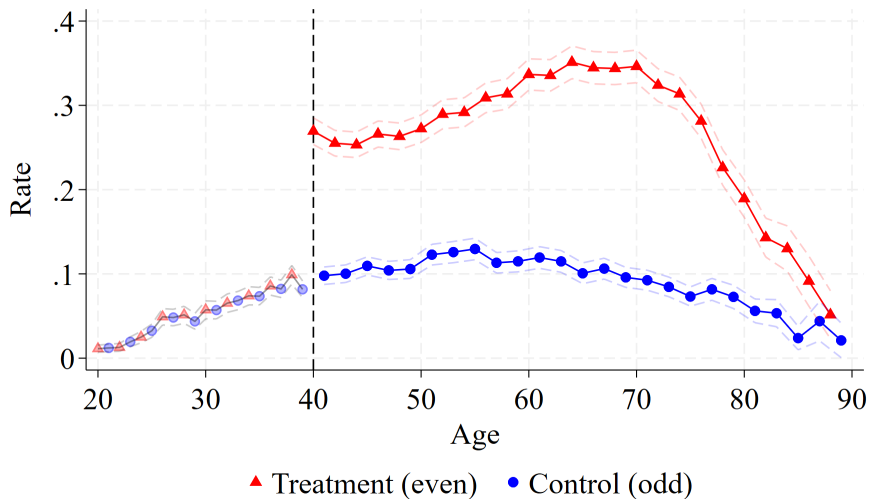
- Effect on take-up**

- Cross spillover

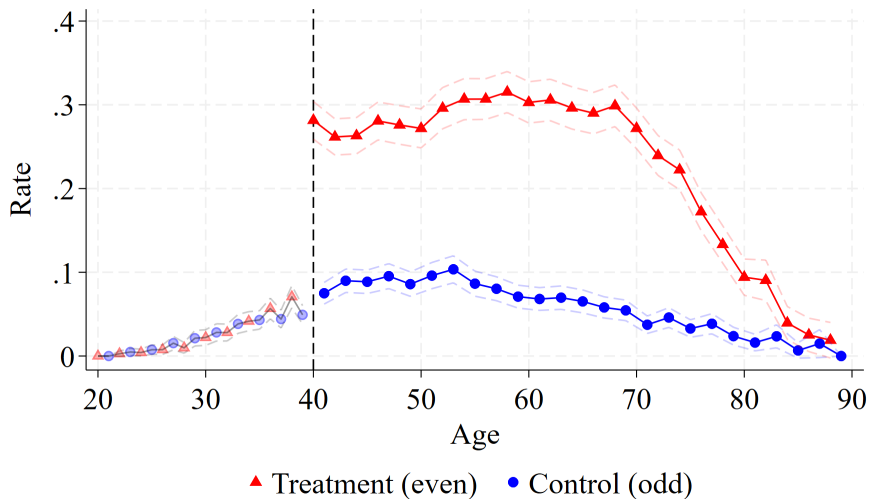
- Selection

- Effect

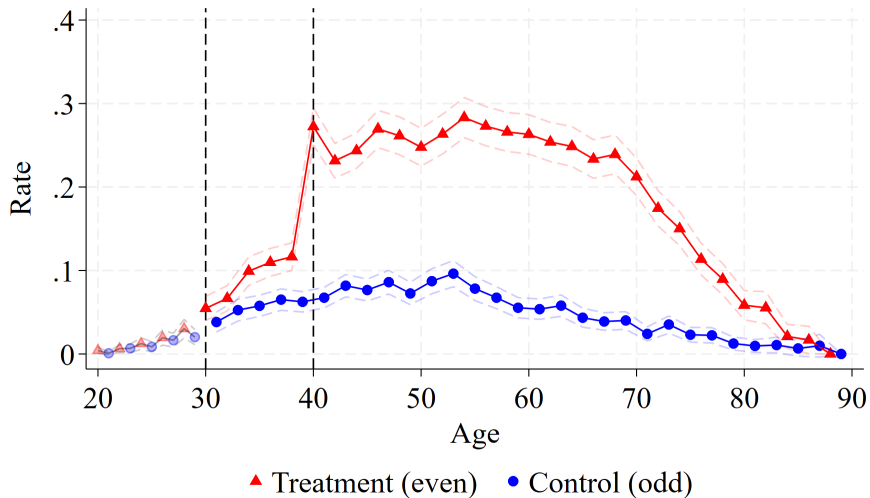
General screening



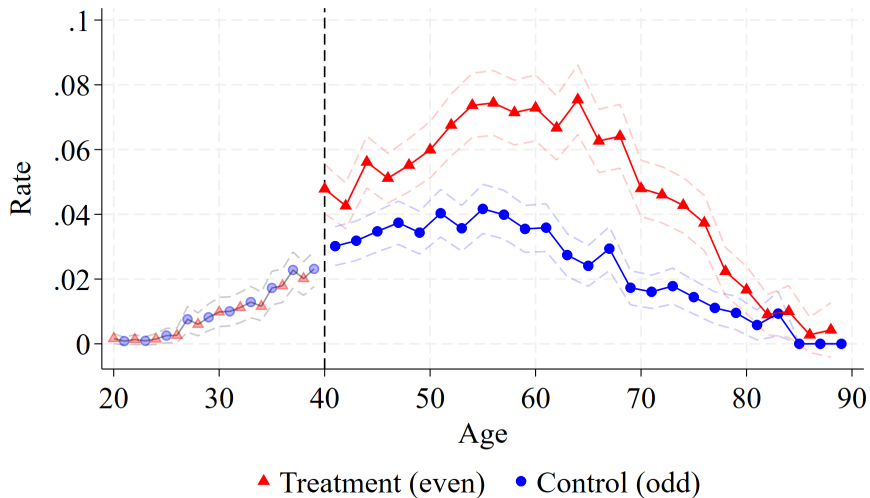
Breast screening



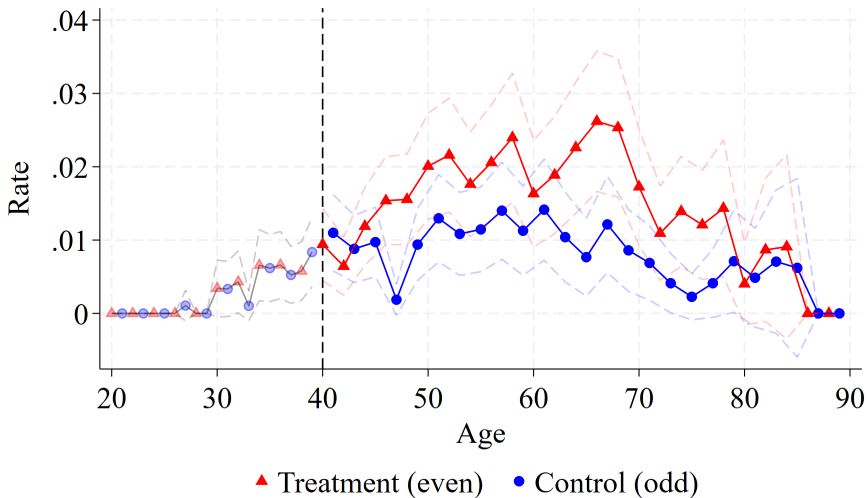
Cervical screening



Liver screening



Prostate screening



Pre-40 and post-40 control participants are similar

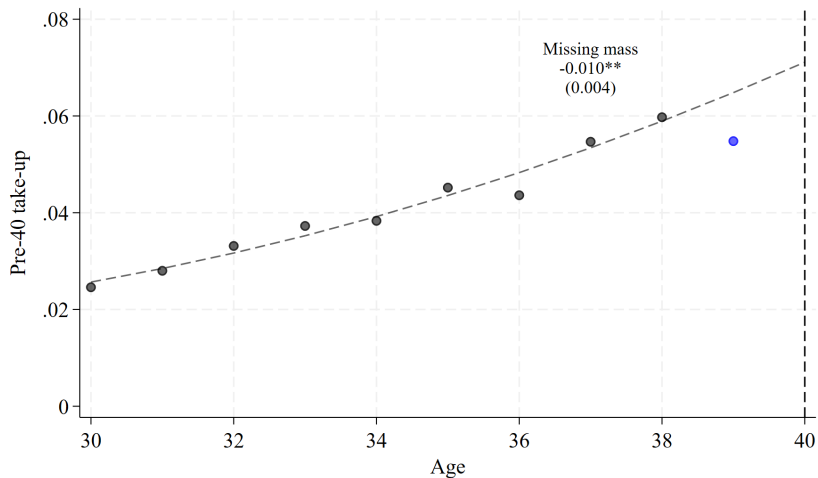
	(1)	(2)	(3)
	Pre-40 always-takers	Post-40 always-takers	Differences
Panel A. Diagnoses			
Stomach	0.132 (0.021)	0.138 (0.021)	0.006 (0.029)
Breast	0.018 (0.012)	0.030 (0.014)	0.012 (0.019)
Panel B. SES			
Individual income	2624 (99)	2898 (143)	274* (157)
Household income	5884 (127)	5955 (152)	70 (183)
Years of education	14.138 (0.102)	14.282 (0.119)	0.144 (0.139)
College graduate	0.388 (0.023)	0.460 (0.025)	0.072** (0.030)
Working status	0.765 (0.019)	0.766 (0.020)	0.000 (0.025)

Pre-40 and post-40 control participants are similar

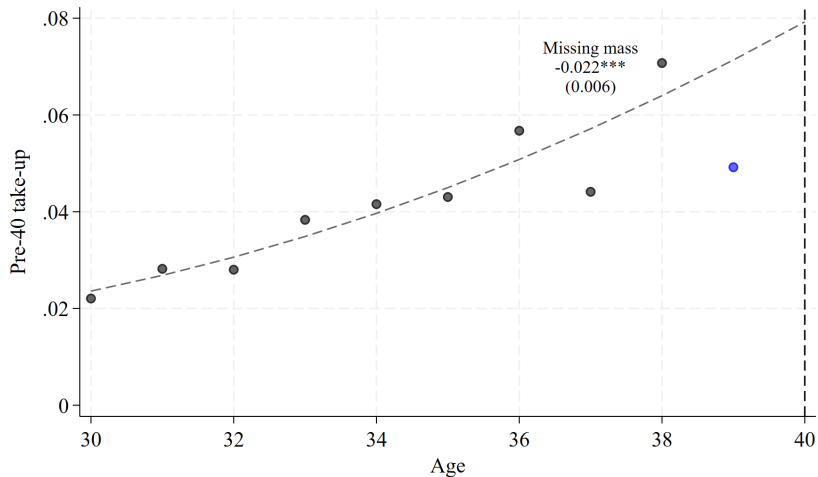
	(1)	(2)	(3)
	Pre-40 always-takers	Post-40 always-takers	Differences
Panel C. Health behaviors			
Current smoker	0.467 (0.038)	0.481 (0.038)	0.014 (0.050)
Everyday smoker	0.446 (0.038)	0.452 (0.038)	0.006 (0.049)
Current drinker	0.892 (0.022)	0.944 (0.019)	0.053* (0.028)
Everyday drinker	0.074 (0.015)	0.076 (0.020)	0.003 (0.024)
Vigorous exercise	0.420 (0.034)	0.443 (0.036)	0.023 (0.048)
Moderate exercise	0.549 (0.034)	0.595 (0.036)	0.046 (0.049)
Walking	0.737 (0.029)	0.764 (0.030)	0.027 (0.041)

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Minimal substitution at age 40 for stomach screening



Minimal substitution at age 40 for breast screening



Arc elasticity comparison

- Arc elasticities from this study: -0.48 to -0.72
- Arc elasticities from RAND HIE: -0.17 to -0.43 ([Newhouse, 1993](#))
- Arc elasticity from another study on screening subsidies: -0.47 ([Kim and Lee, 2017](#))
- Percentage increase in Oregon Medicaid experiment: 45-63% ([Finkelstein et al., 2012](#))

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People get screening earlier before subsidies expire

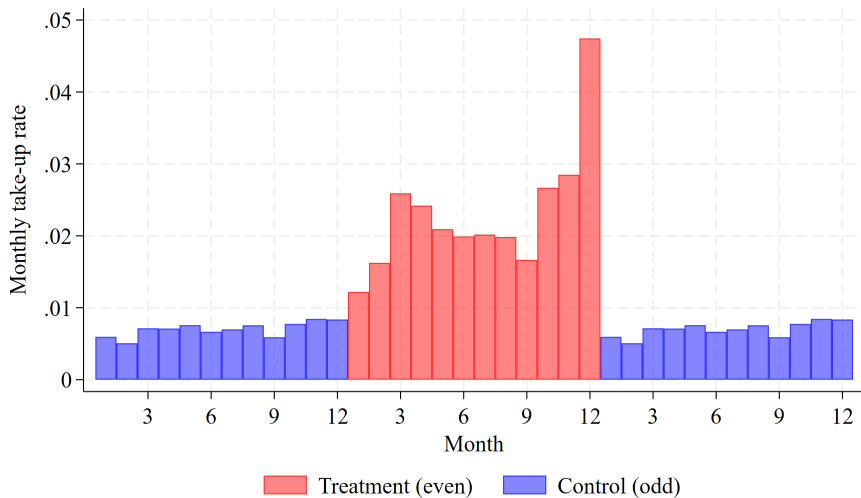


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Multiple screenings on the same day

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Stomach	Breast	Cervical	Liver	Colorectal	Prostate	Lung
$\Pr(\text{general} = 1 \mid \text{screen} = 1)$	0.878	0.861	0.834	0.844	0.799	0.786	0.699
$\Pr(\text{same day} \mid \text{screen} = 1, \text{general} = 1)$	0.964	0.947	0.899	0.948	0.856	0.960	0.937
If not on the same day							
$\Pr(\text{general first} \mid \text{screen} = 1, \text{general} = 1)$	0.028	0.037	0.053	0.035	0.111	0.024	0.043
$\Pr(\text{general later} \mid \text{screen} = 1, \text{general} = 1)$	0.007	0.015	0.032	0.009	0.178	0.004	0.005

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Screening results

- Screening results
 - Find any disease? $\Rightarrow$ Which disease? (ICD-10)
 - Multiple answers allowed
 - Not available for never-takers

Screening	Take-up	Disease diagnosis
Stomach	17.9%	22.8%
Breast	16.4%	2.2%
Cervical	13.9%	6.7%
Colorectal	4.3%	22.3%

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Disease classifications for stomach

- (K29) Gastritis and duodenitis
- (K52) Other noninfective gastroenteritis and colitis
- (K21) Gastro-oesophageal reflux disease
- (K25) Gastric ulcer
- (B98) Helicobacter pylori
- (K31) Other diseases of stomach and duodenum
- (K20) Esophagitis
- (C16) Malignant neoplasm of stomach
- (K26) Duodenal ulcer

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Disease classifications for breast

- (N63) Unspecified lump in breast
- (N64) Other disorders of breast
- (D24) Benign neoplasm of breast
- (N60) Benign mammary dysplasia
- (C50) Malignant neoplasm of breast

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Disease classifications for female genital

- (N76) Other inflammation of vagina and vulva
- (N71) Inflammatory disease of uterus, except cervix
- (N85) Other noninflammatory disorders of uterus, except cervix
- (N83) Noninflammatory disorders of ovary, fallopian tube and broad ligament

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Disease classifications for colon and rectum

- (K63) Other diseases of intestine
- (D12) Benign neoplasm of colon, rectum, anus and anal canal
- (D13) Benign neoplasm of other and ill-defined parts of digestive system
- (R19) Other symptoms and signs involving the digestive system and abdomen
- (C18) Malignant neoplasm of colon

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1. Estimate Always- and Never-takers characteristics

- Individually identifiable always- and never-takers

	Even age (treatment)	Odd age (control)
Always-takers	Yes	Yes
Compliers	Yes	No
Never-takers	No	No

- Estimating equation

$$y_{it} = \beta_0 + \beta_1 \text{treat}_{it} + \beta_2 \text{screen}_{it} + \beta_3 \text{treat}_{it} \times \text{screen}_{it} + \nu_{it} \quad (4)$$

- Average characteristics

- Always-takers: $g_{AT}(y) = \hat{\beta}_0 + \hat{\beta}_2$
- Never-takers: $g_{NT}(y) = \hat{\beta}_0 + \hat{\beta}_1$

2. Back out complier characteristics

- Treated compliers in the treatment group, untreated compliers in the control group

	Even age (treatment)	Odd age (control)
Always-takers	Yes	Yes
Compliers	Yes	No
Never-takers	No	No

- Estimating equation

$$y_{it} = \beta_0 + \beta_1 \text{treat}_{it} + \beta_2 \text{screen}_{it} + \beta_3 \text{treat}_{it} \times \text{screen}_{it} + \nu_{it} \quad (5)$$

- Those getting screened in the treatment group

$$\begin{aligned} g_T(y) &= \frac{\pi_{AT}}{\pi_{AT} + \pi_C} g_{AT}(y) + \frac{\pi_C}{\pi_{AT} + \pi_C} g_C^1(y) \\ &= \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 \end{aligned}$$

- Those not getting screened in the control group

$$\begin{aligned} g_U(y) &= \frac{\pi_{NT}}{\pi_{NT} + \pi_C} g_{NT}(y) + \frac{\pi_C}{\pi_{NT} + \pi_C} g_C^0(y) \\ &= \hat{\beta}_0 \end{aligned}$$

3. Compare compliers to always- and never-takers

- Taking ratios
 - Treated compliers to always-takers: $g_C^1(y)/g_{AT}(y)$
 - Untreated compliers to never-takers: $g_C^0(y)/g_{NT}(y)$
- Why differentiate between treated and untreated compliers?
 - Characteristics in the same year
 - Unclear pre-determined characteristics
 - Difference between treated and untreated complier characteristics = LATE
- Minor details in estimation
 - Age = 60
 - Standard error calculated with bootstrap with clustering at individual level (500 replications)

Stomach diseases and cancer

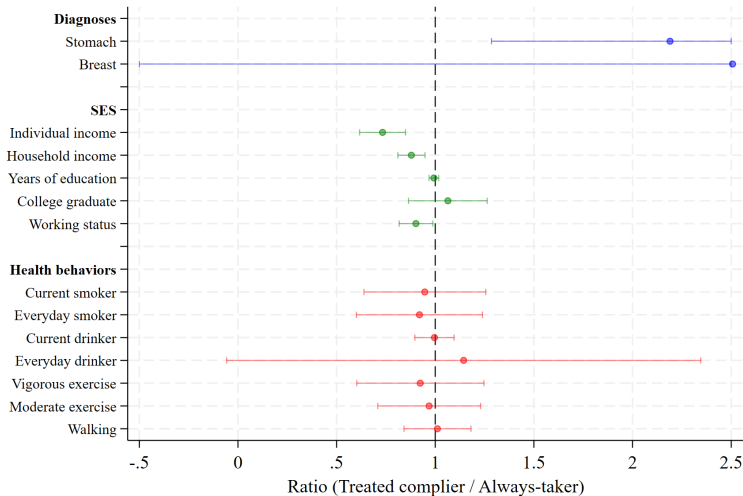
	(1)	(2)	(3)
	Average value		Ratio
	Always-takers	Treated compliers	CP_1 / AT
Helicobacter Pylori	0.009 (0.002)	0.013 (0.002)	1.498 (0.360)
Gastritis	0.123 (0.005)	0.210 (0.006)	1.707*** (0.101)
Stomach ulcer	0.005 (0.001)	0.008 (0.001)	1.444 (0.529)
Stomach cancer	0.001 (0.001)	0.003 (0.001)	1.909 (1.924)
Colon polyp	0.070 (0.007)	0.114 (0.016)	1.628* (0.359)

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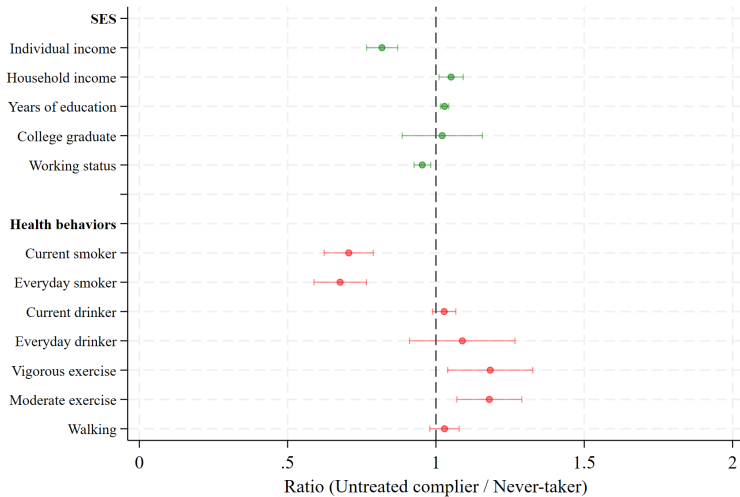
Stomach diseases and cancer

- Unlike in US, stomach cancer is common in South Korea
 - % of all new cancer cases: 1.3% (US 2024) 11.5% (SK 2019)
 - % of all cancer deaths: 1.8% (US 2024) 9.4% (SK 2019)
- There is high returns to early detection of stomach cancer
 - 5-year relative survival rate
 - Stomach: localized (97.4%) → distant (6.6%)
 - Breast: localized (98.9%) → distant (45.2%)
- Stomach screening (upper endoscopy)
 - Cause: Helicobacter Pylori ([Uemura et al., 2001](#))
 - Symptoms: inflammation (gastritis), ulcer (peptic ulcer), cancer
 - Diagnosed through endoscopy
 - Treatment: 2 weeks of antibiotics and an acid-reducing proton pump inhibitor

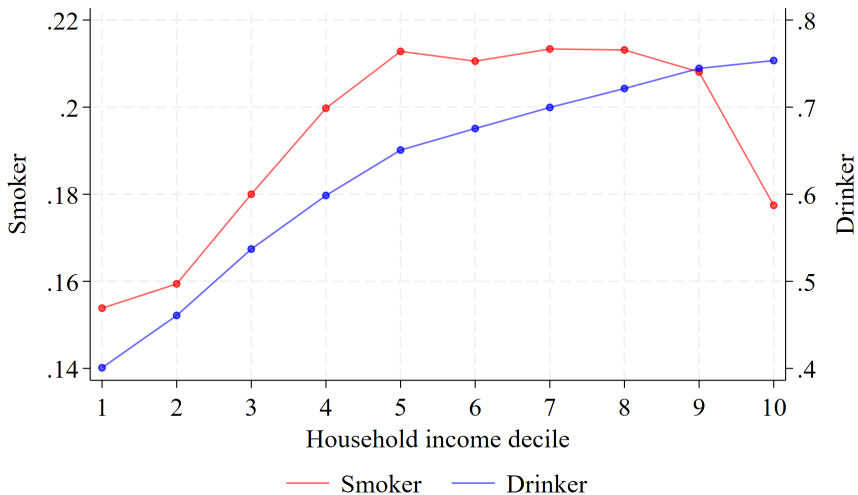
Compliers vs Always-takers at the age 40 cutoff



Compliers vs Never-takers



Household income and health behaviors



Compliers among non-working population

	(1)	(2)	(3)	(4)	(5)	(6)
	Average value				Ratio	
	Always-takers	Treated compliers	Untreated compliers	Never-takers	CP_1/AT	CP_0/NT
Panel A. Diagnoses						
Stomach	0.171 (0.011)	0.283 (0.012)	- -	- -	1.654*** (0.144)	- -
Breast	0.018 (0.005)	0.027 (0.005)	- -	- -	1.552 (0.897)	- -
Panel B. SES						
Individual income	166 (18)	159 (19)	166 (20)	171 (17)	0.958 (0.112)	0.969 (0.099)
Household income	4381 (134)	4131 (131)	4186 (138)	3728 (107)	0.943* (0.031)	1.123*** (0.028)
Years of education	10.363 (0.147)	10.430 (0.154)	10.396 (0.151)	9.652 (0.126)	1.007 (0.015)	1.077*** (0.012)
College graduate	0.128 (0.013)	0.113 (0.013)	0.124 (0.013)	0.096 (0.010)	0.881 (0.109)	1.293*** (0.111)

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Use of medical tests

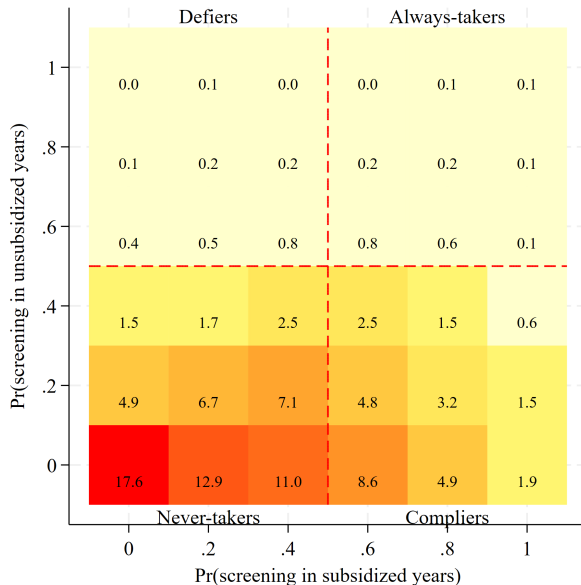
	(1)	(2)	(3)
	Average value		Ratio
	Always-takers	Treated Compliers	CP_1/AT
Panel A. Tests covered by NHIS			
Blood/urine/stool/X-ray	0.795 (0.005)	0.885 (0.006)	1.114*** (0.012)
Endoscopy	0.812 (0.006)	0.834 (0.006)	1.027** (0.011)
Biopsy	0.025 (0.002)	0.030 (0.002)	1.181 (0.154)
Panel B. Tests not covered by NHIS			
Sonogram	0.319 (0.007)	0.274 (0.007)	0.857*** (0.031)
CT	0.042 (0.003)	0.016 (0.002)	0.370*** (0.070)
MRI	0.010 (0.001)	0.006 (0.001)	0.608** (0.183)
PET	0.002 (0.001)	-0.000 (0.000)	-0.213*** (0.264)
EEG	0.002 (0.001)	0.002 (0.001)	0.755 (0.586)
EKG	0.159 (0.005)	0.126 (0.005)	0.790*** (0.048)
Bone density	0.034 (0.003)	0.028 (0.003)	0.844 (0.123)

Using 10 years of panel to define compliance groups

- Analytical sample ($N = 5,701$)
 - Balanced sample without attrition for 10 years (2009-2018)
 - First year age ≥ 40
- Probability of screening in subsidized and unsubsidized years

$$Pr(\text{screen_even}) = \frac{1}{5} \times \sum_{k=1}^5 \mathbb{1}\{\text{screen}_{ik} = 1\}, \quad k \text{ even} \quad (6)$$

Bivariate distribution of screening in treatment and control years



Compliers have poorer health

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Always	Compliers	Defiers	Never	Compliers / Always	Compliers / Defiers	Compliers / Never
Panel A. Diagnoses							
Stomach	0.181 (0.018)	0.256 (0.007)	0.189 (0.020)	- -	1.413*** (0.143)	1.355** (0.146)	- -
Breast	0.015 (0.007)	0.020 (0.003)	0.022 (0.011)	- -	1.303 (0.653)	0.885 (0.452)	- -
Cervical	0.061 (0.015)	0.058 (0.005)	0.058 (0.024)	- -	0.950 (0.249)	0.997 (0.417)	- -
Colorectal	0.183 (0.030)	0.260 (0.013)	0.212 (0.032)	- -	1.423* (0.248)	1.226 (0.197)	- -
Share	0.022	0.294	0.025	0.659			
Pr(even/odd or odd/even)	0.510	0.483	0.590	0.500			

Compliers have poorer health

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Always	Compliers	Defiers	Never	Compliers / Always	Compliers / Defiers	Compliers / Never
Panel A. Diagnoses							
Stomach	0.181 (0.018)	0.256 (0.007)	0.189 (0.020)	- -	1.413*** (0.143)	1.355** (0.146)	- -
Breast	0.015 (0.007)	0.020 (0.003)	0.022 (0.011)	- -	1.303 (0.653)	0.885 (0.452)	- -
Cervical	0.061 (0.015)	0.058 (0.005)	0.058 (0.024)	- -	0.950 (0.249)	0.997 (0.417)	- -
Colorectal	0.183 (0.030)	0.260 (0.013)	0.212 (0.032)	- -	1.423* (0.248)	1.226 (0.197)	- -
Share	0.022	0.294	0.025	0.659			
Pr(even/odd or odd/even)	0.510	0.483	0.590	0.500			

- Compliers have poorer health than always-takers

Compliers have lower SES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Always	Compliers	Defiers	Never	Compliers / Always	Compliers / Defiers	Compliers / Never
Panel B. SES							
Individual income	2688 (247)	1043 (37)	2405 (222)	1288 (28)	0.388*** (0.038)	0.434*** (0.043)	0.810*** (0.034)
Household income	6093 (308)	3999 (67)	5587 (284)	3764 (46)	0.656*** (0.035)	0.716*** (0.038)	1.063*** (0.022)
Years of education	12.184 (0.312)	9.876 (0.098)	11.486 (0.349)	9.585 (0.075)	0.811*** (0.022)	0.860*** (0.027)	1.030** (0.013)
College graduate	0.270 (0.039)	0.123 (0.008)	0.291 (0.038)	0.142 (0.006)	0.456*** (0.072)	0.423*** (0.062)	0.869** (0.066)
Working status	0.753 (0.030)	0.560 (0.010)	0.733 (0.032)	0.603 (0.007)	0.744*** (0.033)	0.764*** (0.036)	0.930*** (0.020)

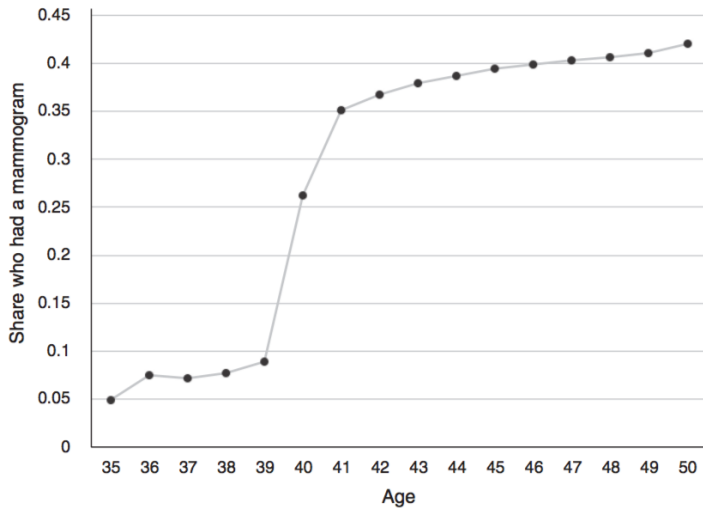
- Compliers have lower SES than always-takers

Compliers show better health behaviors than never-takers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Always	Compliers	Defiers	Never	Compliers / Always	Compliers / Defiers	Compliers / Never
Panel C. Health behaviors							
Current smoker	0.272 (0.043)	0.268 (0.016)	0.381 (0.047)	0.399 (0.010)	0.984 (0.165)	0.702*** (0.096)	0.670*** (0.043)
Everyday smoker	0.260 (0.042)	0.257 (0.015)	0.350 (0.046)	0.386 (0.010)	0.991 (0.170)	0.735** (0.106)	0.667*** (0.044)
Current drinker	0.828 (0.042)	0.775 (0.014)	0.856 (0.031)	0.763 (0.008)	0.935 (0.050)	0.905*** (0.036)	1.016 (0.021)
Everyday drinker	0.111 (0.027)	0.141 (0.010)	0.175 (0.033)	0.151 (0.006)	1.274 (0.329)	0.807 (0.161)	0.933 (0.078)
Vigorous exercise	0.374 (0.032)	0.300 (0.010)	0.340 (0.028)	0.278 (0.006)	0.802*** (0.074)	0.883 (0.079)	1.080* (0.043)
Moderate exercise	0.526 (0.031)	0.474 (0.010)	0.525 (0.031)	0.419 (0.006)	0.902* (0.056)	0.903* (0.056)	1.133*** (0.029)
Walking	0.830 (0.023)	0.812 (0.007)	0.818 (0.019)	0.780 (0.005)	0.979 (0.029)	0.993 (0.025)	1.042*** (0.011)

- Compliers show better health behaviors than never-takers

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Panel A. Share of mammograms that are true positive and false positive

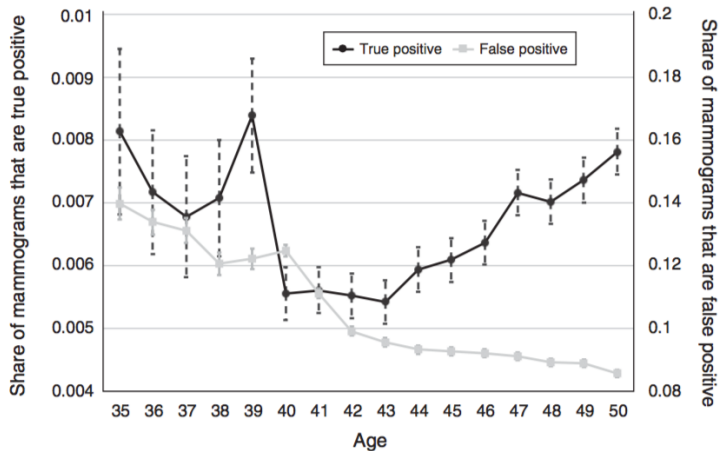


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Back-up slides

- Research question

- Institutional background

- Identification and data

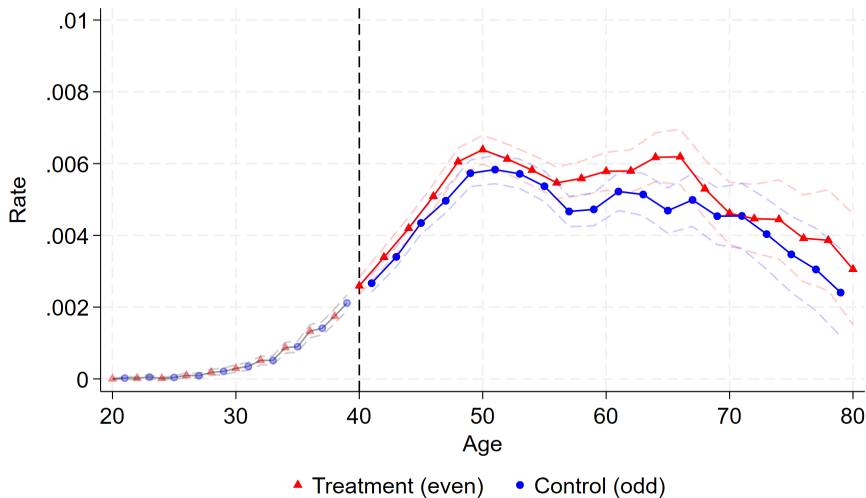
- Effect on take-up

- Cross spillover

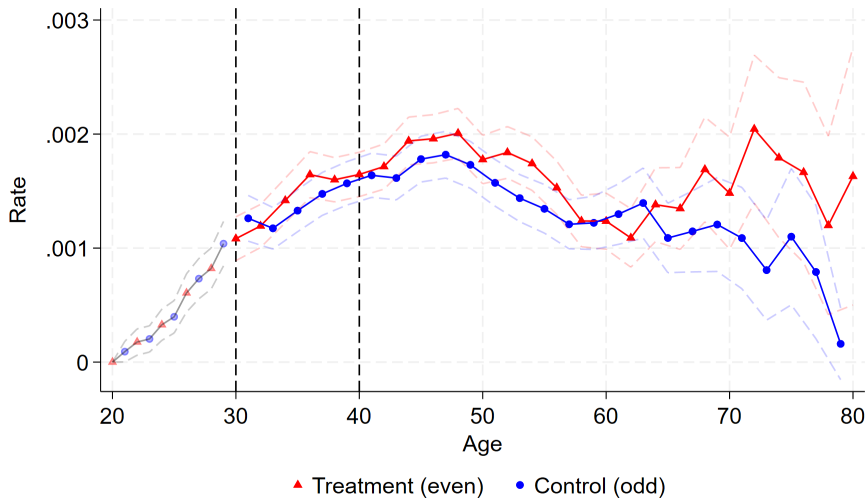
- Selection

- Effect

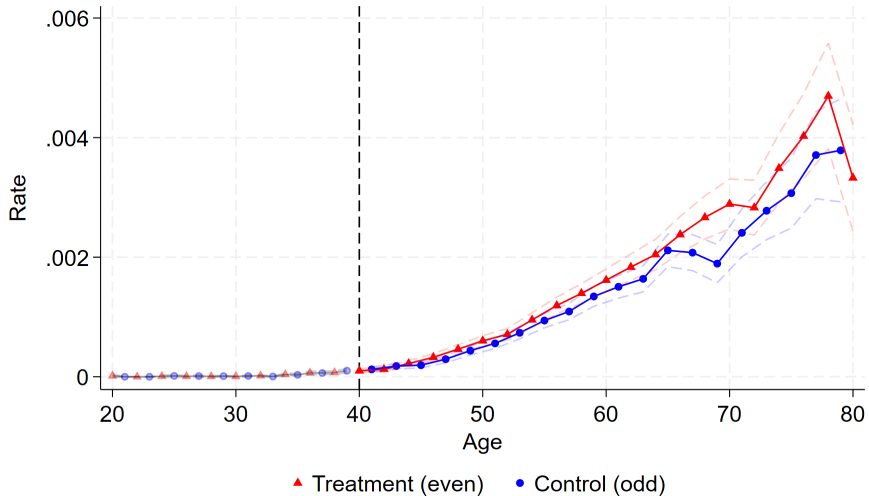
More breast cancer diagnoses in the treatment group



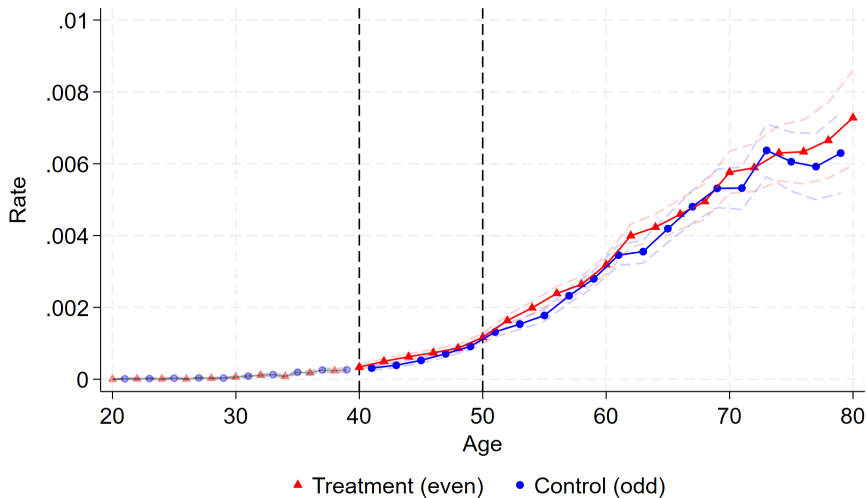
More cervical cancer diagnoses in the treatment group



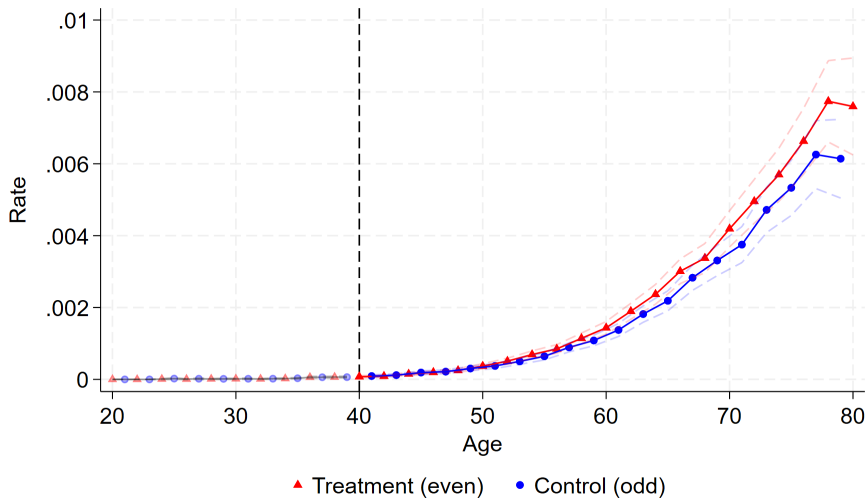
More liver cancer diagnoses in the treatment group



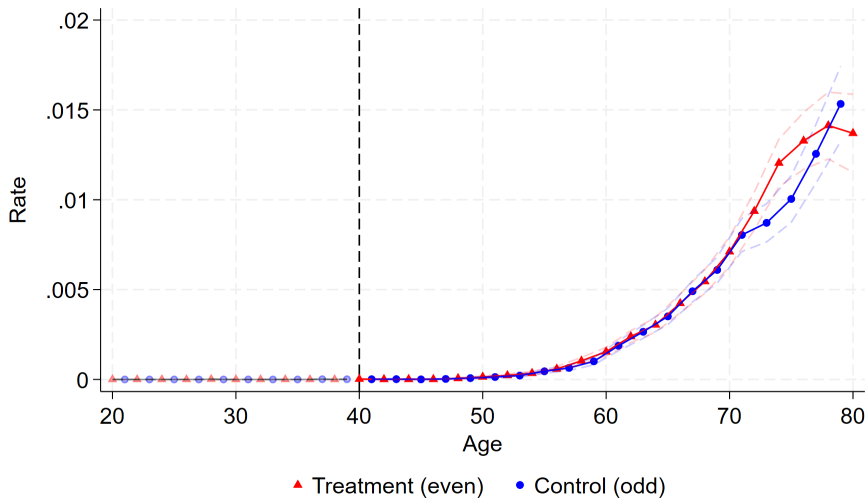
More colorectal cancer diagnoses in the treatment group



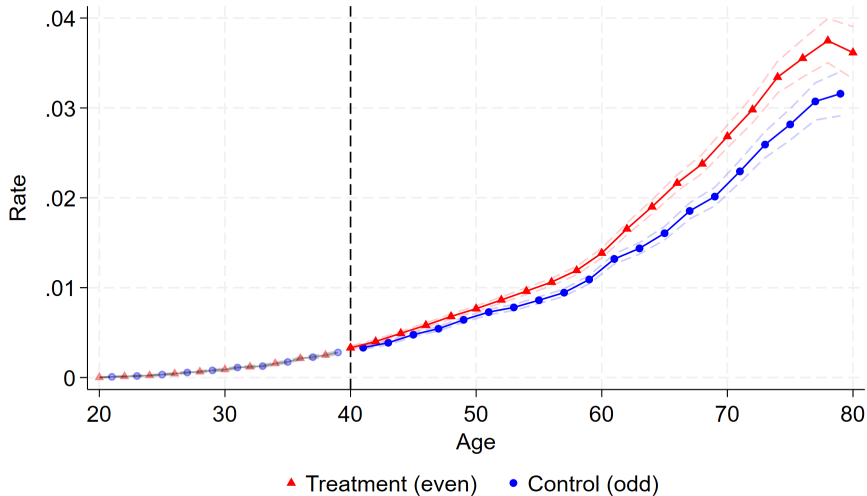
More lung cancer diagnoses in the treatment group



'Slightly' more prostate cancer diagnoses in the treatment group



Overall, screening leads to more cancer diagnoses



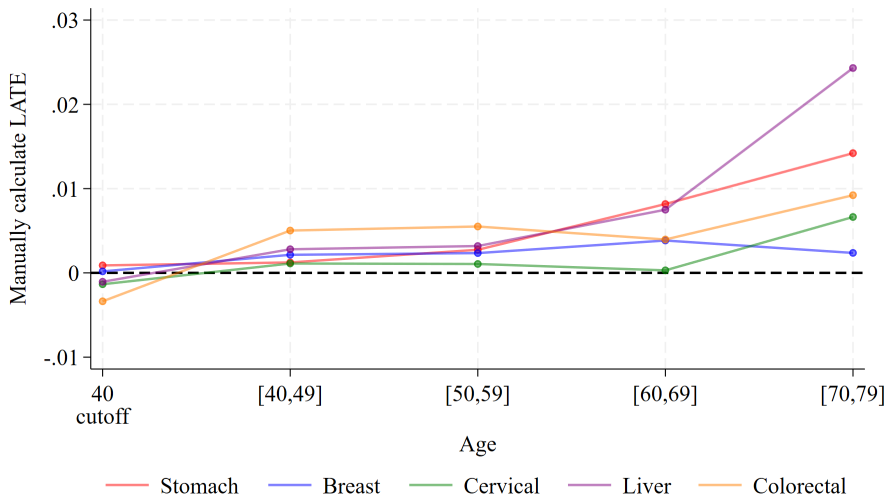
LATE comparison with other studies

- Local Average Treatment Effect = ITT (insurance claims) / First stage (survey)
 - Cancer detection rate among compliers
- LATE for breast screening: 0.0026
 - Organized breast screening program in Europe: 0.0010 (Guthmuller et al., 2023)
 - US mammogram recommendation at age 40: 0.0019 (Kadiyala and Strumpf, 2016)
 - Another mammogram subsidy in Korea: 0.0025 (Kim and Lee, 2017)
- LATE for stomach screening: 0.0041
 - Another stomach screening subsidy in Korea: 0.0025 (Kim and Lee, 2017)

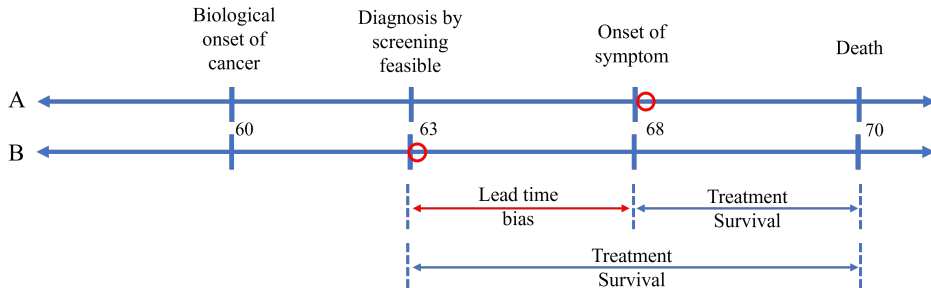
Diagnoses increase for non-biennially subsidized cancer screenings

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
Liver cancer	90	13*** (4)	16	7,449,256
invasive	90	13*** (4)	15	7,449,256
in-situ	0.5	0.2 (0.2)	43	7,449,256
Colorectal cancer	190	18*** (5)	10	7,449,256
invasive	170	17*** (5)	10	7,449,256
in-situ	23	3* (1)	12	7,449,256
Lung cancer	90	14*** (4)	15	7,449,256
invasive	90	14*** (4)	15	7,449,256
in-situ	0.7	-0.1 (0.2)	-20	7,449,256
Prostate cancer	160	10 (7)	6	3,945,600
invasive	150	10 (7)	6	3,945,600
in-situ	1.1	-0.09 (0.44)	-9	3,945,600

Manually calculated LATE estimates by age



Lead time bias in survival rate analysis



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Screening prevents future cancers by treating cancer precursors

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
Helicobacter pylori	17	3*** (1)	15	8,673,954
Polypectomy	1,174	724*** (9)	62	8,673,954

Screening increases medication use for osteoporosis

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
Tuberculosis	195	1 (2)	0.5	8,673,954
Osteoporosis	2,613	106*** (7)	4	8,673,954

Screening increases medication use for high blood pressure

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
High blood pressure	0.23884	0.00046*** (0.00013)	0.193	8,673,954
ARBs (Angiotensin II Receptor Blockers)	0.10179	0.00038*** (0.00009)	0.378	8,673,954
Calcium Channel Blockers (CCBs)	0.09779	0.00021** (0.00009)	0.218	8,673,954
Diuretics (Water Pills)	0.07136	-0.00003 (0.00010)	-0.046	8,673,954
Beta-Blockers	0.04289	0.00005 (0.00008)	0.121	8,673,954
ACE Inhibitors	0.01357	-0.00003 (0.00004)	-0.233	8,673,954
Aldosterone Antagonists	0.00970	-0.00007 (0.00004)	-0.705	8,673,954
Alpha-Blockers	0.00768	0.00001 (0.00004)	0.109	8,673,954

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Screening increases medication use for diabetes

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
Diabetes	0.08918	0.00015** (0.00006)	0.171	8,673,954
Biguanides	0.06832	0.00011* (0.00007)	0.165	8,673,954
Sulfonylureas	0.04318	-0.00006 (0.00006)	-0.139	8,673,954
DPP-4 Inhibitors	0.01836	0.00005 (0.00004)	0.298	8,673,954
Alpha-Glucosidase Inhibitors	0.00432	0.00001 (0.00002)	0.195	8,673,954
Meglitinides	0.00174	-0.00001 (0.00002)	-0.303	8,673,954
Insulin (Various Types)	0.00165	0.00001 (0.00001)	0.615	8,673,954
GLP-1 Receptor Agonists	0.00014	0.00000 (0.00001)	3.127	8,673,954
SGLT2 Inhibitors	0.00202	-0.00000 (0.00002)	-0.204	8,673,954
Thiazolidinediones (TZDs)	0.00671	-0.00002 (0.00003)	-0.299	8,673,954

Screening increases medication use for high cholesterol

	(1)	(2)	(3)	(4)
	Control group mean	ITT	Percent relative to control	N
High cholesterol	0.13901	0.00419*** (0.00011)	3.014	8,673,954
Statins	0.13359	0.00405*** (0.00011)	3.033	8,673,954
Fibrates	0.00752	0.00016*** (0.00004)	2.076	8,673,954
Ezetimibe	0.00194	0.00006*** (0.00002)	3.131	8,673,954

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Robustness with different age controls: first stage

	(1)	(2)	(3)	(4)
	General	Stomach	Breast	Cervical
Panel A. Linear splines of age				
Interval 3	0.187*** (0.003)	0.190*** (0.003)	0.191*** (0.004)	0.144*** (0.003)
Interval 5	0.187*** (0.003)	0.190*** (0.003)	0.191*** (0.004)	0.145*** (0.003)
Interval 7	0.187*** (0.003)	0.190*** (0.003)	0.191*** (0.004)	0.144*** (0.003)
Panel B. Linear splines with 5 years interval plus additional covariates				
Full controls	0.187*** (0.003)	0.190*** (0.003)	0.191*** (0.004)	0.145*** (0.003)
Individual FE	0.189*** (0.003)	0.191*** (0.003)	0.192*** (0.004)	0.146*** (0.003)

Robustness

Robustness with different age controls: cross spillover

	(1)	(2)	(3)	(4)
	Liver	Colorectal	Prostate	Lung
Panel A. Linear splines of age				
Interval 3	0.027*** (0.001)	0.033*** (0.001)	0.007*** (0.001)	0.0062*** (0.0007)
Interval 5	0.027*** (0.001)	0.033*** (0.001)	0.007*** (0.001)	0.0062*** (0.0007)
Interval 7	0.027*** (0.001)	0.033*** (0.001)	0.007*** (0.001)	0.0062*** (0.0007)
Panel B. Linear splines with 5 years interval plus additional covariates				
Full controls	0.027*** (0.001)	0.033*** (0.001)	0.007*** (0.001)	0.0062*** (0.0007)
Individual FE	0.028*** (0.001)	0.033*** (0.001)	0.007*** (0.001)	0.0063*** (0.0007)

Robustness

Robustness: selection analysis with different age controls

	(1)	(2)	(3)	(4)	(5)	(6)
	Interval 3		Interval 5		Interval 7	
	CP_1/AT	CP_0/NT	CP_1/AT	CP_0/NT	CP_1/AT	CP_0/NT
Panel A. Diagnoses						
Stomach	1.679*** (0.083)	- -	1.685*** (0.084)	- -	1.684*** (0.083)	- -
Breast	1.234 (0.338)	- -	1.228 (0.354)	- -	1.212 (0.328)	- -
Cervical	0.911 (0.153)	- -	0.906 (0.154)	- -	0.909 (0.153)	- -
Colorectal	1.189 (0.159)	- -	1.190 (0.162)	- -	1.190 (0.161)	- -
Panel B. SES						
Individual income	0.591*** (0.029)	0.818*** (0.027)	0.591*** (0.029)	0.815*** (0.027)	0.585*** (0.029)	0.816*** (0.027)
Household income	0.875*** (0.018)	1.050** (0.022)	0.875*** (0.018)	1.047** (0.021)	0.872*** (0.018)	1.055*** (0.021)
Years of education	0.972*** (0.009)	1.031*** (0.007)	0.971*** (0.009)	1.031*** (0.007)	0.971*** (0.009)	1.031*** (0.007)
College graduate	0.612*** (0.055)	1.041 (0.067)	0.601*** (0.057)	1.026 (0.069)	0.609*** (0.055)	1.032 (0.067)
Working status	0.864*** (0.016)	0.959*** (0.015)	0.865*** (0.016)	0.956*** (0.014)	0.865*** (0.016)	0.955*** (0.015)

Robustness: selection analysis with different age controls

	(1)	(2)	(3)	(4)	(5)	(6)
	Interval 3		Interval 5		Interval 7	
	CP_1/AT	CP_0/NT	CP_1/AT	CP_0/NT	CP_1/AT	CP_0/NT
Panel C. Health behaviors						
Current smoker	0.938 (0.065)	0.710*** (0.041)	0.934 (0.067)	0.711*** (0.042)	0.935 (0.065)	0.714*** (0.041)
Everyday smoker	0.956 (0.068)	0.683*** (0.044)	0.952 (0.070)	0.683*** (0.045)	0.954 (0.068)	0.684*** (0.044)
Current drinker	0.987 (0.019)	1.031 (0.020)	0.987 (0.019)	1.030 (0.020)	0.987 (0.019)	1.030 (0.020)
Everyday drinker	0.920 (0.095)	1.069 (0.091)	0.915 (0.095)	1.084 (0.090)	0.920 (0.092)	1.080 (0.089)
Vigorous exercise	0.906* (0.049)	1.208*** (0.075)	0.911* (0.049)	1.189*** (0.073)	0.910* (0.049)	1.198*** (0.074)
Moderate exercise	0.932* (0.038)	1.177*** (0.057)	0.930* (0.038)	1.180*** (0.056)	0.931* (0.038)	1.181*** (0.057)
Walking	0.999 (0.021)	1.018 (0.025)	0.994 (0.021)	1.030 (0.025)	0.994 (0.021)	1.030 (0.025)

Bounding estimates: first stage

	(1)	(2)	(3)
	Age $\in [39, 89]$	Age $\in [40, 89]$	Age $\in [41, 89]$
General	0.187*** (0.003)	0.186*** (0.003)	0.188*** (0.003)
Stomach	0.191*** (0.003)	0.189*** (0.003)	0.190*** (0.003)
Breast	0.192*** (0.004)	0.191*** (0.004)	0.190*** (0.004)
Cervical	0.165*** (0.003)	0.165*** (0.003)	0.162*** (0.003)
N	110121	107183	104153

Robustness

Bounding estimates: cross spillover

	(1)	(2)	(3)
	Age $\in [39, 89]$	Age $\in [40, 89]$	Age $\in [41, 89]$
Liver	0.027*** (0.001)	0.026*** (0.001)	0.027*** (0.001)
Colorectal	0.033*** (0.001)	0.033*** (0.001)	0.034*** (0.001)
Lung	0.0061*** (0.0007)	0.0061*** (0.0007)	0.0064*** (0.0007)
Prostate	0.007*** (0.001)	0.007*** (0.001)	0.008*** (0.001)
N	110121	107183	104153

Robustness

RD at cutoff: first stage

	(1)	(2)	(3)	(4)
	General	Stomach	Breast	Cervical
Pre-cutoff take-up ($\hat{\alpha}_0$)	0.088*** (0.006)	0.063*** (0.005)	0.056*** (0.007)	0.026*** (0.006)
N	23,623	23,623	12,077	7,797
Sample age range	[36, 43]	[36, 43]	[36, 43]	[26, 33]
Subsidy starting age	40	40	40	30

Robustness

RD at cutoff: first stage

	(1)	(2)	(3)	(4)
	General	Stomach	Breast	Cervical
Pre-cutoff take-up ($\hat{\alpha}_0$)	0.088*** (0.006)	0.063*** (0.005)	0.056*** (0.007)	0.026*** (0.006)
Change in treatment group ($\hat{\alpha}_2 + \hat{\alpha}_4$)	0.181*** (0.010)	0.201*** (0.009)	0.225*** (0.013)	0.029*** (0.009)
N	23,623	23,623	12,077	7,797
Sample age range	[36, 43]	[36, 43]	[36, 43]	[26, 33]
Subsidy starting age	40	40	40	30

Robustness

RD at cutoff: first stage

	(1)	(2)	(3)	(4)
	General	Stomach	Breast	Cervical
Pre-cutoff take-up ($\hat{\alpha}_0$)	0.088*** (0.006)	0.063*** (0.005)	0.056*** (0.007)	0.026*** (0.006)
Change in control group ($\hat{\alpha}_2$)	0.009 (0.010)	0.011 (0.009)	0.011 (0.012)	0.006 (0.011)
Change in treatment group ($\hat{\alpha}_2 + \hat{\alpha}_4$)	0.181*** (0.010)	0.201*** (0.009)	0.225*** (0.013)	0.029*** (0.009)
N	23,623	23,623	12,077	7,797
Sample age range	[36, 43]	[36, 43]	[36, 43]	[26, 33]
Subsidy starting age	40	40	40	30

Robustness

RD at cutoff: first stage

	(1)	(2)	(3)	(4)
	General	Stomach	Breast	Cervical
Pre-cutoff take-up ($\hat{\alpha}_0$)	0.088*** (0.006)	0.063*** (0.005)	0.056*** (0.007)	0.026*** (0.006)
Change in control group ($\hat{\alpha}_2$)	0.009 (0.010)	0.011 (0.009)	0.011 (0.012)	0.006 (0.011)
Change in treatment group ($\hat{\alpha}_2 + \hat{\alpha}_4$)	0.181*** (0.010)	0.201*** (0.009)	0.225*** (0.013)	0.029*** (0.009)
Post-cutoff treatment – Post-cutoff control ($\hat{\alpha}_4$)	0.173*** (0.011)	0.189*** (0.011)	0.214*** (0.015)	0.024* (0.012)
N	23,623	23,623	12,077	7,797
Sample age range	[36, 43]	[36, 43]	[36, 43]	[26, 33]
Subsidy starting age	40	40	40	30

Robustness

RD at cutoff: cross spillover

	(1)	(2)
	Cervical	Colorectal
Panel A. Odd age group		
Pre-40 take-up	0.070*** (0.006)	0.019*** (0.002)
Change at age 40	0.003 (0.008)	0.001 (0.003)
Panel B. Even age group		
Pre-40 take-up	0.138*** (0.009)	0.017*** (0.002)
Change at age 40	0.114*** (0.012)	0.023*** (0.004)
N	27,714	54,642
Adj R^2	0.062	0.008
Even group % increase	82	139
Sample age range	[30, 49]	[30, 49]
Subsidy starting age	30	50

Robustness

RD at cutoff: selection into screening

	(1)	(2)	(3)	(4)	(5)	(6)
	Average value				Ratio	
	Always-takers	Treated compliers	Untreated compliers	Never-takers	CP_1/AT	CP_0/NT
Panel A. Diagnoses						
Stomach	0.132 (0.021)	0.288 (0.018)	- -	- -	2.190*** (0.462)	- -
Breast	0.018 (0.012)	0.044 (0.010)	- -	- -	2.507 (750.215)	- -
Cervical	0.167 (0.024)	0.067 (0.023)	- -	- -	0.398*** (0.179)	- -
Colorectal	0.161 (0.039)	0.213 (0.058)	- -	- -	1.320 (0.695)	- -
Panel B. SES						
Individual income	2624 (99)	1923 (116)	1667 (188)	2263 (45)	0.733*** (0.059)	0.737*** (0.090)
Household income	5884 (127)	5174 (139)	4018 (276)	4740 (57)	0.879*** (0.035)	0.848** (0.063)
Years of education	14.138 (0.102)	14.039 (0.116)	13.549 (0.218)	13.642 (0.051)	0.993 (0.012)	0.993 (0.018)
College graduate	0.388 (0.023)	0.413 (0.025)	0.281 (0.045)	0.323 (0.010)	1.064 (0.102)	0.870 (0.150)
Working status	0.765 (0.019)	0.691 (0.023)	0.615 (0.042)	0.766 (0.008)	0.902** (0.044)	0.803*** (0.058)

RD at cutoff: selection into screening

	(1)	(2)	(3)	(4)	(5)	(6)
	Average value				Ratio	
	Always-takers	Treated compliers	Untreated compliers	Never-takers	CP_1/AT	CP_0/NT
Panel C. Health behaviors						
Current smoker	0.467 (0.038)	0.442 (0.048)	0.714 (0.093)	0.573 (0.015)	0.947 (0.158)	1.247 (0.174)
Everyday smoker	0.446 (0.038)	0.410 (0.048)	0.681 (0.092)	0.545 (0.015)	0.920 (0.163)	1.249 (0.182)
Current drinker	0.892 (0.022)	0.888 (0.031)	0.833 (0.059)	0.903 (0.009)	0.996 (0.051)	0.923 (0.069)
Everyday drinker	0.074 (0.015)	0.084 (0.026)	0.106 (0.050)	0.075 (0.008)	1.144 (0.613)	1.407 (0.770)
Vigorous exercise	0.420 (0.034)	0.389 (0.047)	0.255 (0.088)	0.355 (0.013)	0.924 (0.164)	0.719 (0.265)
Moderate exercise	0.549 (0.034)	0.532 (0.046)	0.406 (0.095)	0.462 (0.013)	0.969 (0.133)	0.880 (0.220)
Walking	0.737 (0.029)	0.746 (0.042)	0.808 (0.087)	0.729 (0.012)	1.011 (0.087)	1.108 (0.131)

Robustness

Heterogeneous effects on cancer diagnoses

